Xiemin (Simon) Wen

Email: x5wen@uwaterloo.ca Phone: +1 (226) 234-8126 Personal Website: xieminwen.com

EDUCATION

University of Waterloo — Waterloo, Ontario

Major: **Computer Science** (GPA: 3.7/4.0) — Sept 2023 - Present

PROFESSIONAL EXPERIENCE

QA Automation Intern – Tangam System (Software company for casino analytics) — May 2026 - Aug 2026

- Developed and maintained 200+ automated end-to-end test cases across 70+ product tickets using Cypress, JavaScript, and Cucumber for a casino analytics platform.
- Automated regression testing for user-facing web workflows, validating feature behavior, data presentation, and cross-page functionality through reusable E2E test scenarios.
- Designed and validated 10+ automation scenarios across four major product areas during a focused testing cycle, expanding regression coverage for high-priority product workflows.

QA Analyst Intern – Propel Holdings (AI-powered fintech company) — Sep 2025 - Dec 2025

- Performed SQL-driven backend validation across 100+ QA tickets, verifying customer and loan-application data across databases, APIs, UI output, and business-rule logic.
- Developed reusable SQL queries to reconcile backend records with API responses and user-facing data, identifying inconsistencies across financial product workflows.
- Validated data integrity and application behavior across database, backend, and frontend layers for production financial systems and lending workflows.

Project Lead - Microsoft & University of Waterloo — May 2024 - Aug 2024

- Led a 4-person engineering team in developing an AI-powered investment insights platform that generated personalized guidance from users' financial profiles, investment goals, and market interests.
- Co-developed the AI, backend, and database layers using Python, Microsoft Azure, and Azure SQL to support investment insights, suggested allocations, and financial-report analysis.
- Owned project planning, technical coordination, team meetings, and final delivery, earning an Outstanding overall performance rating for the work term.

PROJECTS

AI Artifact Store — July 2026 - Aug 2026

(C++20, Distributed Systems, PostgreSQL, Boost.Asio/Beast, SHA-256)

- Built a C++20 distributed content-addressed storage (CAS) system using SHA-256, FastCDC/fixed-size chunking, PostgreSQL metadata, and Boost.Asio/Beast networking with resumable artifact transfers.
- Implemented multi-node replication and repair, resumable push/pull, garbage collection, lifecycle management, and end-to-end integrity verification across storage nodes.
- Validated storage correctness with 415 automated tests (409 GoogleTest cases + 6 process-level E2E tests) and benchmarked FastCDC at ~95.8% chunk reuse on shifted-content workloads versus 0% for fixed-size chunking.

Adaptive AI Inference Runtime — July 2026 – Aug 2026

(C++20, Concurrency, llama.cpp, Boost.Asio/Beast, CMake)

- Built a multi-threaded C++20 inference runtime with bounded admission, workload-aware scheduling, multi-worker routing, worker-local dynamic batching, model residency, eviction, and overload backpressure.
- Integrated llama.cpp for Qwen3.5-0.8B Q4_0 GGUF inference with multi-sequence infer_batch, Apple Metal offload, and CPU fallback; validated with 152 CTest entries plus ASan/UBSan and TSan coverage.
- Benchmarked real-model HTTP serving on Apple M4 Pro; at concurrency 4, adaptive batching raised median throughput from 2.17 to 4.35 req/s (+101%) and cut p95 end-to-end latency from 2.85 s to 2.11 s (-26.1%).

TECHNICAL SKILLS

C++, Python, JavaScript, SQL, PostgreSQL, CMake, llama.cpp, Distributed Systems, Concurrency, AI Infrastructure